



# Introduction

Section 1 Definition and history of Fluid Mechanics

Section 2 Overview and applications of Fluid Mechanics

Section 3 Research methods in Fluid Mechanics





# Introduction

## Section 1 Definition and history of Fluid Mechanics



### I, Definition of Fluid Mechanics

### II, History of Fluid Mechanics





# Section 1 Definition and history of Fluid Mechanics

## 1, Definition of Fluid Mechanics

Fluid Mechanics is an independent branch of Physics. It is defined as the study of fluids, their laws of motion, and their practical application.

### Theory

- Hydrostatics: static equilibrium
- Fluid dynamics: forces, motion, balance laws

### Applications

- Hydraulic engineering: hydraulics, pipes, channels, etc
- Mechanical engineering: machines, metallurgy, industry, etc
- Civil engineering: construction, bridges, flood control, etc





# Introduction

## Section 1 Definition and history of Fluid Mechanics

### I, Definition of Fluid Mechanics

### II, History of Fluid Mechanics





## Section 2 History of Fluid Mechanics

- Beginning — **245 BC**, Greek philosopher **Archimedes** wrote “On Floating Bodies”
- Formation — **1687 Newton** wrote “Mathematical Principles of Natural Philosophy”

- 3 steps of development
1. Bernoulli, Euler: hydrodynamics of inviscid fluids
  2. Navier, Stokes: hydrodynamics of viscous fluids, beginning experimental research
  3. End of 19<sup>th</sup> century: equal emphasis on theory and experiments



# Major events:

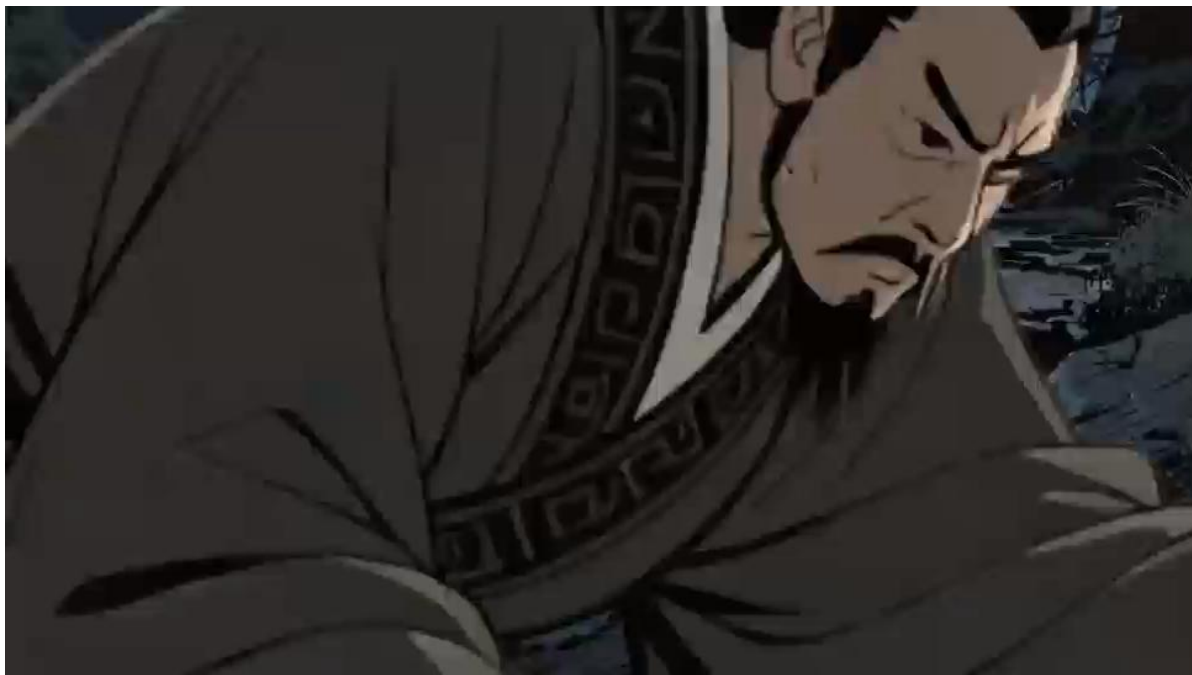


- In **1738**, the Swiss mathematician **Bernoulli** proposed the Bernoulli equation in his famous book "Fluid Dynamics".
- In **1755**, **Euler** put forward the concept of ideal fluid in "General Principles of Fluid Motion" and established the energy conservation equation, thus proposing an analytical description of fluid motion and the concept of velocity potential.
- In **1781**, **Lagrange** introduced the concept of stream function.
- In **1823**, French engineer **Navier** and in **1845**, British mathematician and physicist **Stokes** proposed the famous Navier-Stokes equations.
- In **1876**, **Reynolds** discovered two types of fluid flow: laminar and turbulent.
- In **1858**, **Helmholtz** pointed out properties of vortices in ideal fluids and the theory of vortex motion; in **1887**, he proposed the theory of flow around a body.
- At the end of the **19th century**, the similarity theory was proposed, combining experiment and theoretical analysis.
- In **1904**, **Prandtl** proposed the boundary layer theory.
- After the **1960s**, **computational fluid dynamics** has evolved rapidly, and the meaning of "Fluid Mechanics" has been continuously enriched.



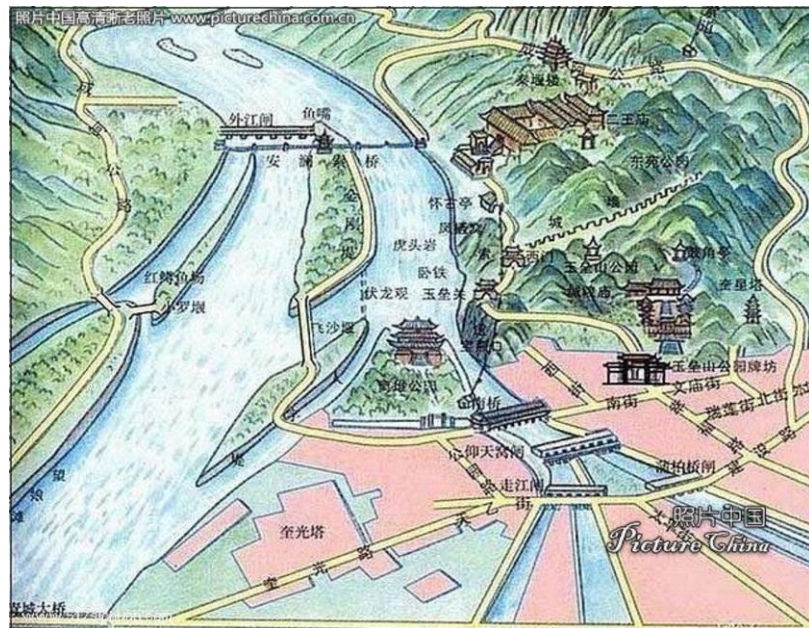
## Li Bing

**Li Bing (302-235 BC)** was a Chinese hydraulic engineer and politician of the Warring States period. He served the state of **Qin** as an administrator and is revered for his work on the **Dujiangyan River Control System**, which both controlled flooding and provided irrigation water year-round, greatly increasing the productivity of the valley. Li Bing became a cultural icon, known as the vanquisher of the River God and is compared to the Great Yu. Dujiangyan is **still in use today** and is listed as a UNESCO World Heritage Site.



# Dujiangyan

**Dujiangyan** was originally constructed around 256 BC by the State of Qin as **an irrigation and flood control project**, it is still in use today. The system's infrastructure develops on the Min River (Minjiang), the longest tributary of the Yangtze. The area is in the west part of the Chengdu Plain, between the Sichuan Basin and the Tibetan Plateau. Originally, the Min would rush down from the Min Mountains and slow down abruptly after reaching the Chengdu Plain, filling the watercourse with silt, thus making the nearby areas extremely prone to floods. King Zhao of Qin commissioned the project, and the construction of the Dujiangyan harnessed the river using a new method of channeling and dividing the water rather than simply damming it. The water management scheme is still in use today to irrigate over 5,300 km<sup>2</sup> of land in the region. The Dujiangyan, the Zhengguo Canal in Shaanxi and the Lingqu Canal in Guangxi are collectively known as the "three great hydraulic engineering projects of the Qin."





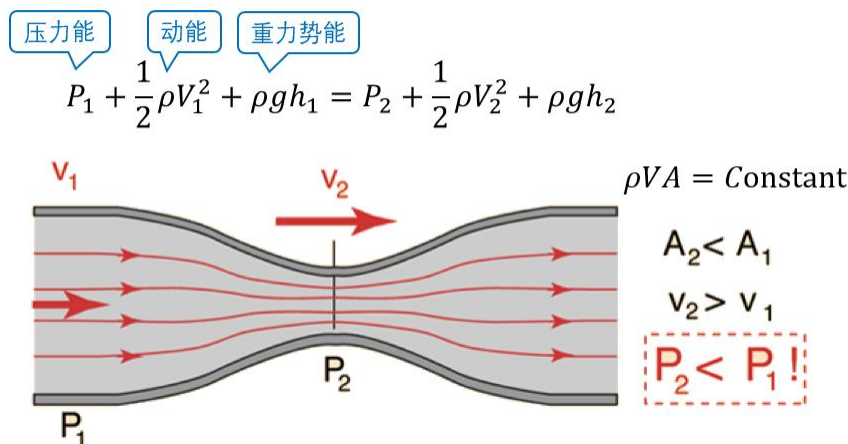
# Daniel Bernoulli



**Daniel Bernoulli (1700-1782)** is a Swiss scientist who taught at the Petersburg Academy of Sciences in Russia. He made significant contributions to fluid mechanics, gas dynamics, differential equations and probability theory, and is the founder of fluid mechanics theory. Bernoulli is best known for his book "Hydrodynamics" (1738), in which he proposed the **laws of conservation of energy** in ideal fluids (incompressible, viscous fluids). This theorem and the corresponding formula are called Bernoulli's theorem and **Bernoulli's formula**. He also has many works on solid mechanics.



Daniel Bernoulli  
1700 ~ 1782



# Leonardo da Vinci

**Leonardo da Vinci** (1452-1519) is an Italian Renaissance **artist**, natural **scientist**, **engineer**, and **founder of theoretical mechanics**. He made important contributions to the **formation of classical theories of hydraulics and fluid mechanics**. His philosophical thought is close to materialism, he believes that everything in the world is subject to the objective law of necessity, knowledge originates from feeling, and he also points out the importance of theoretical generalization. He emphasized that mathematics and mechanics are the foundations of the natural sciences. **He made many important discoveries in the fields of hydraulic engineering, civil engineering, and mechanical engineering**. Leonardo da Vinci's studies of mechanics were not limited to theory. He also used mechanics and mechanical principles to design many machines and instruments, and participated in the design and construction of canals, hydraulic engineering and construction projects. **He first designed a flying machine in 1493 by studying the motion of bird wings**. He wrote many important manuscripts on hydraulics, which were published posthumously under the title "**Motion and Measurement of Water**".



# Leonardo da Vinci



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# Leonhard Euler

**Leonhard Euler** (1707-1783) is a Swiss mathematician, mechanic, astronomer, physicist, founder of variational method, pioneer of complex function theory, and **founder of theoretical fluid mechanics**. Euler was a professor at the Petersburg Academy of Sciences and one of the founders of the Berlin Academy of Sciences. He is the founder of rigid body mechanics and fluid mechanics, and the founder of the stability theory of elastic systems. He believed that the differential equations of particle dynamics could be applied to liquids (1750). He has used **two methods to describe the motion of fluids, namely, describing the fluid velocity field in terms of a fixed point in space (1755) and in terms of a determined fluid particle (1759)**. The former is called **Euler's method**, and the latter is called **Lagrange's method**. Euler laid the theoretical foundation for the dynamics of ideal fluid, and proposed the mass (1752) and momentum conservation (1755) equations. Euler also wrote many works on solid mechanics. Euler has more than 800 monographs and papers.







## History of hydraulic engineering in China:

- The story of "Da Yu's water control" more than 4,000 years ago.
- The Qin Dynasty built the three major hydraulic engineering projects (Dujiangyan, Zhengguo Canal, Ling Canal) from 256 BC to 210 BC.
- About 2500 years ago, the water clock was invented in China.
- During the Yongzheng period of the Qing Dynasty, He Mengyao proposed that the flow rate is equal to the cross-sectional area of the water multiplied by the average flow velocity of the cross-section.
- The North-South Grand Canal, completed in the Sui Dynasty (587-610).
- The Zhaozhou Stone Arch Bridge (595-605), built during the Sui Dynasty on the Xianhe River in Jizhong – world's oldest arch bridge





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## Section 2 Overview and applications of Fluid Mechanics



1, Nature and purpose of the course

2, Overview of applications of Fluid Mechanics

3, Basic course requirements

4, Learning problems and countermeasures





## Section 2 Overview and applications of Fluid Mechanics

### 1, Nature and purpose of the course

- Nature: A compulsory professional basic course for civil engineering and hydraulic engineering majors
- Purpose: Learn theory and how to apply
- Prerequisites: Physics, Theoretical Mechanics, Mechanics of Materials
- Follow-ups: hydrology, soil mechanics, hydraulics, engineering geology, etc.
- Others: a. Compulsory subjects for the registered engineer examination; b. One of the compulsory or elective subjects in the Graduate Entrance Examination.







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## 2, Overview of applications of Fluid Mechanics

1. **Civil and construction engineering:** foundation drainage, roadbed drainage, groundwater infiltration, anti-seepage stabilization, dam construction;
2. **Urban engineering:** bridge and culvert aperture design, water supply and drainage, pipe network calculation, pump station and water tower design, etc;
3. **Urban flood control projects:** dikes, dams, flood control gates, etc.;
4. **Building environment:** heating, ventilation and air conditioning design, etc;
5. **Hydraulic Engineering:** Wider application and deeper course content, follow-up course "Engineering Hydraulics".





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### 3, Basic course requirements

- Obtain comprehensive theoretical basis, including:
  1. Master the **basic theoretical concepts** of Fluid Mechanics
  2. Become familiar with **flow analysis**, combining dimensional analysis and experiment, and with the method of solving simple plain potential flow
  3. Master the **laws of fluid motion, energy conversation**, and **head loss**, and become familiar with flow resistance
- Obtain the ability to analyze and discuss certain **practical problems**, including:
  1. Calculation of hydraulic load
  2. Calculation of flow capacity of pipes, channels; calculation of seepage
  3. Analysis and calculation of head loss
- Master methods of **measuring** water level, pressure, velocity and flow







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## 4, Learning problems and countermeasures

- Concepts too abstract  $\xleftrightarrow{\text{countermeasures}}$  Check alternative sources (books, internet, Wikipedia), ask for assistance (e.g., Thomas)
- Mathematical derivations too difficult  $\leftarrow$  try to understand step by step (**do not memorize**), derivations techniques often repeat themselves
- Many kinds of PDEs  $\leftarrow$  emphasis is on physical meaning, application, and main solution methods





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# Introduction

## Section 3 Research methods in Fluid Mechanics



1, Theoretical methods

2, Experimental methods

3, Numerical methods





## Section 3 Research methods in Fluid Mechanics

### 1, Theoretical methods

- **Pure theoretical approach:** Build theoretical models and apply mathematical methods to obtain theoretical results
- **Combine theory with experiments:** Build theoretical models and determine certain parameter values from measurements

- **Main theorems used in theoretical approach:**

(1) Mass conservation:  $\frac{dm}{dt} = 0$

(2) Momentum conservation:  $\sum \vec{F} = \frac{d(m\vec{u})}{dt}$

(3) Energy conservation: Kinetic energy + pressure energy + potential energy + energy loss = constant





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1, Theoretical methods



2, Experimental methods

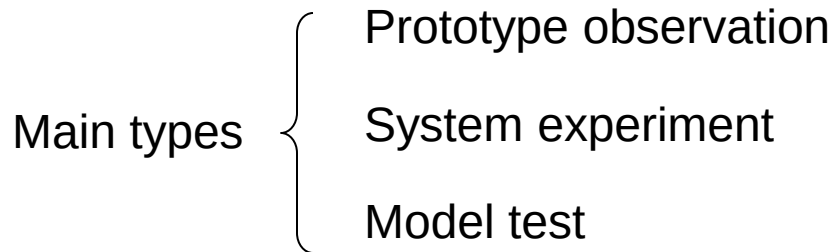
3, Numerical methods



## 2, Experimental methods

**Applied Fluid Mechanics** is a basic subject closely combining theory and practice. Many of its **practical formulas and coefficients** are derived from experiments. So far, many problems in engineering, even if they can be solved by modern theoretical analysis and numerical calculation, have to be corrected by experimental verification.

### 1, Experimental research type:



2. The basic method of **experimental research** (detailed in Chapter 5): Similarity theory, such as **Re** similarity or **Fr** similarity between prototype and model

Reynolds number (Re):  $Re = \frac{dv}{\nu}$

Froude number (Fr):  $Fr = \frac{v}{\sqrt{gh}}$





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### 3, Numerical methods

**Numerical methods** are based on discretizing **theoretical Fluid Mechanic equations** using various **discretization methods** (finite difference method, finite element method, etc.). **Numerical calculations** and numerical **experiments** are then carried out through computers to obtain the flow field in space and time. In the past 20 to 30 years, many numerical models to solve Fluid Mechanics equations have been developed, giving rise to a new discipline - **Computational Fluid Dynamics**.

